

Lesson 1: Rectilinear motion

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1. A mountain biker starts from rest at a trailhead. As she descends, her GPS records her displacement in meters as a function of time as

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$$x = 0.5t^3 + t^2 + 2t$$

Determine her position, velocity, and acceleration when $t=4$ seconds.

2. The brakes of a car are applied, causing it to slow down at a rate of 8 ft/s^2 . Knowing that the car stops in 400 ft, determine (a) how fast the car was traveling immediately before the brakes were applied, (b) the time required for the car to stop.

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3. The acceleration of a drone is given as

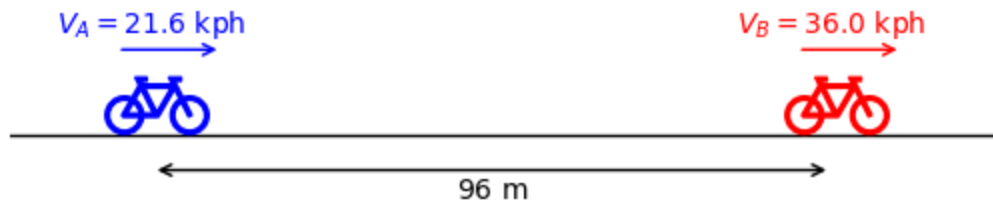
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$$a = 16 - 3t^2$$

where a is in ft/s^2 and t is in seconds. The drone starts at $t=0$ with $v=0$ at $x=100$ feet. Determine (a) the time when the velocity is again zero, (b) the position and the velocity when $t=6$ s, (c) the total distance traveled by the drone from $t=0$ to $t=6$ s.

4. A sprinter runs a 100-m race. For the first 8 seconds the runner accelerates uniformly and covers a distance of 32 meters. The runner then maintains a constant velocity for the remaining 68 meters. Determine the runner's a) acceleration, b) final velocity, c) finishing time.

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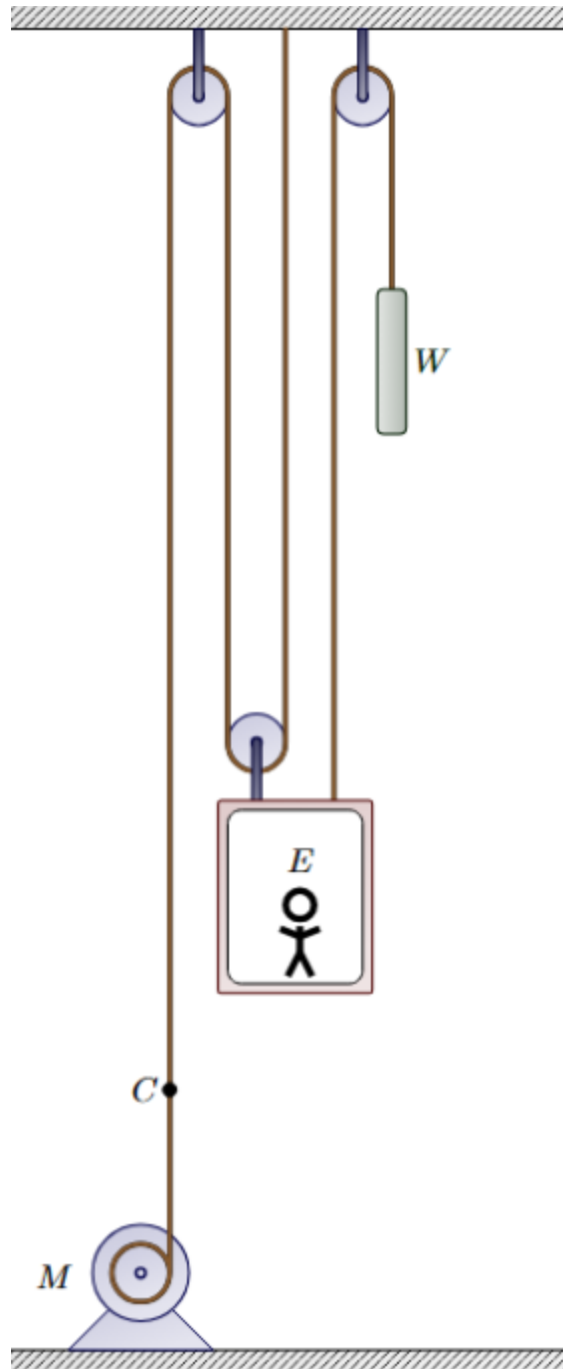


A cyclist is trying to close a gap to a rider ahead. At $t=0$ the cyclists have the positions and speeds as shown. Knowing that cyclist A accelerates at a rate of 3 m/s^2 and cyclist B is decelerating at 1 m/s^2 , determine (a) when and where cyclist A will overtake cyclist B , (b) the speed of each cyclist at that time.



6. A schematic for an elevator is shown in the image. At some instant the elevator is moving downward with a constant velocity of 4 m/s. For the following questions, please report the magnitude (absolute value) for each result.

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What is the speed of point C on the cable?

What is the velocity of the counterweight, W ?

What is V_{CE} , the relative velocity of the cable C with respect to the elevator E ?

What is V_{WE} , the relative velocity of the counterweight W with respect to the elevator E ?
